Development of a Computational Phantom of the Vertebral Column for Dosimetric Investigations

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1. INTRODUCTION

- Spinal tumors present significant therapeutic challenges due to their critical location and proximity to the spinal cord. Brachytherapy with phosphorus-32 (^{32}P), a short-range beta emitter, emerges as a promising alternative, as it allows for high doses to the tumor volume with reduced risk to adjacent tissues.

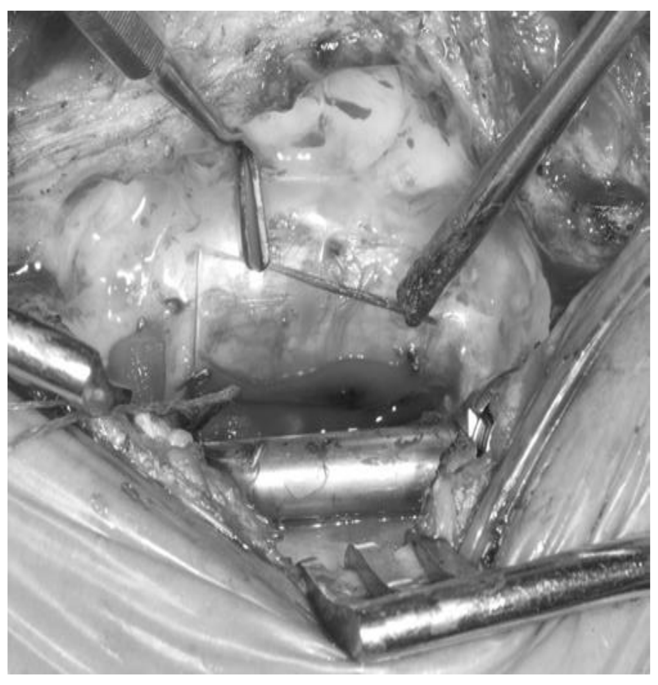

Figure 1: Flexible ^{32}P source
Source: Tong, 2014

- However, the anatomical complexity of the spine makes accurate dose estimation difficult. In this context, computational phantoms derived from CT images enable more realistic Monte Carlo simulations.
- This work presents the development of a spinal column phantom for the dosimetric assessment of ^{32}P, aiming for safer and more effective treatments.

2. MATERIALS AND METHODS

SPINAL CT (The Cancer Imaging Archive)

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Segmentations – Open-source nnU-net algorithm

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Manual Refinament - Blender

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Python-based algorithm for Voxel Conversion (MCNP simulations)

3. RESULTS

- The main result of this work is the generation of a high-fidelity three-dimensional (3D) computational phantom of the vertebral column, segmented from computed tomography (CT) images. The final model incorporates the key anatomical structures relevant for dosimetric studies.

Figure 2: 3D computational phantom of the vertebral column. The segmented model includes a simulated tumor volume in the T12 vertebra, the other vertebrae, and the spinal cord. This realistic geometry is used as input for Monte Carlo dosimetry simulations

4. CONCLUSION

- The development of a computational spinal phantom provides a detailed anatomical representation that is crucial for accurate dosimetric studies of ^{32}P brachytherapy. By enabling realistic Monte Carlo simulations, this approach contributes to a better understanding of dose distribution in complex anatomical regions, supporting safer and more effective treatment planning.

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